AI SYSTEM BRIEF & ARCHITECTURE SPECIFICATION

Account Lookup with Context

Platform: n8n Orchestration | **Pattern:** Linear AI-Supported Enrichment | **Repository:** `account-lookup-with-context/`

1. Executive Overview

The **Account Lookup with Context** workflow is a linear, AI-supported enrichment pipeline built in n8n. Rather than relying on an autonomous multi-step loop, this deterministic workflow follows a predictable sequence: retrieving structured account records from a central database, enriching the payload with an LLM-generated executive summary, preserving the underlying source context, and delivering a ready-to-read briefing directly to Slack.

2. Core Business Problem

> **Operational Bottleneck:** Before customer check-ins, renewal calls, or escalation reviews, account managers and support leads frequently spend 10–15 minutes manually querying CRM tables, reviewing support logs, and synthesizing account history—slowing response times and creating inconsistent pre-meeting visibility.

3. Architecture & Execution Flow

Stage	n8n Node / Component	Functional Responsibility
1. Trigger & Ingestion	`Manual Trigger` → `Test Request`	Captures the target account identifier or briefing request payload.
2. Database Lookup	`Search records` (Airtable / CRM)	Queries the central account database to retrieve structured company records, metrics, and status fields.
3. State Preservation	`Carry Account Context`	Explicitly maps and retains key raw account fields so source data remains accessible downstream alongside AI output.
4. AI Synthesis	`Draft Summary` + `OpenRouter Chat Model`	Passes retrieved account data to an LLM to generate a concise, scannable executive brief.
5. Consolidation	`Combine Summary and Context`	Merges the generated narrative summary with verified account metadata into a single structured payload.
6. Team Delivery	`Send a message` (Slack)	Dispatches the formatted briefing straight to the designated team Slack channel.

4. Architectural Design Rationale (Automation vs. Agent)

Because the data retrieval path is predictable and executes the exact same sequence on every run, this solution deliberately implements a **deterministic AI-supported workflow** rather than an autonomous agent loop. This architectural choice ensures lower token expenditure, faster execution latency, and reliable, repeatable formatting.

5. Business Impact

- **Zero-Latency Pre-Call Prep:** Compresses manual CRM research into an automated, on-demand Slack digest ready seconds before client engagements.
- **Grounded & Verifiable Outputs:** Preserving raw account context alongside the AI summary prevents hallucinated metrics and allows teams to verify key figures at a glance.
- **Cost-Predictable Execution:** Single-pass LLM summarization eliminates runaway loop risk and keeps API overhead minimal.